Wireless Communication

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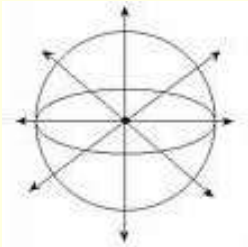
Antenna Concepts

- Antenna is an electrical conductor or system of conductors that radiates or collects electromagnetic energy to/from space.
- Antenna is a passive device. So the power radiated by the antenna is always less than the power entering from the transmitter.
- Antenna is a reciprocal device. So the same antenna can be used for both transmitting and receiving.

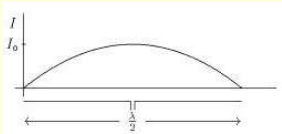
Types of Antenna

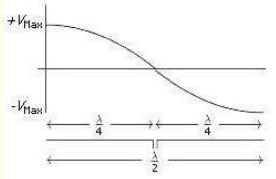
- Isotropic Antenna:
  - An isotropic antenna is a point in space that radiates power in all directions equally.
  - They are just used for theoretical studies. No existence in practice.

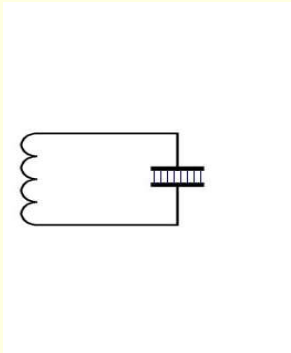


Types of Antenna

- Dipole Antenna:
  - Half-wave dipole antenna is a simple and practical antenna which is in common use.







  - Length of dipole:

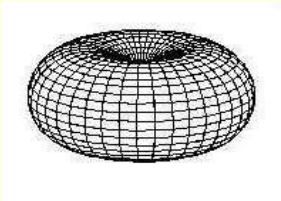
    Free space wave length is

    $\lambda = c / f$

    There is a 5% gap in between. So

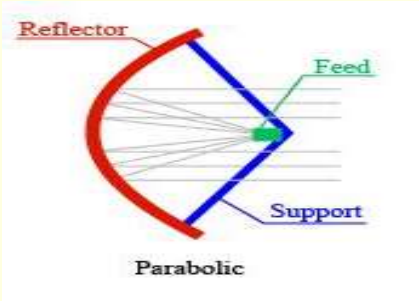
    $L = 0.95 \times 0.5 \times c/f$

    $c = 3 \times 10^8$ m/s



Types of Antenna

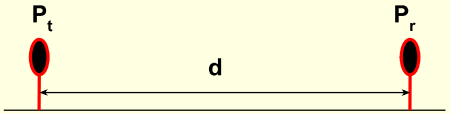
- Parabolic Antenna:
  - A **parabolic antenna** is a high-gain reflector antenna used for radio, television and data communications, and also for radiolocation (radar)



Antenna Parameters

- Power Density:
  $P_{DI} = P_t / (4\pi r^2)$ P_t □ transmitted power
- Antenna Gain:
  $G_t = P_{DR} / P_{DI}$ P_{DR} □ *Power Density of real antenna in a real direction*
  P_{DI} □ *Power density of Isotropic Antenna.*
  Thus $P_{DR} = P_t G_t / (4\pi r^2)$
- Effective Area:
  $A_e = P_A / P_D$ PA □ Absorbed Power, PD □ Power density
  $= (\lambda^2 / 4\pi)\, G$
- Antenna Efficiency:
  $\eta = A_e / A_{phy}$ η *is 0.35 to 0.55*
- Path Loss:
  $P_r = A_e.P_t.G_t/(4\pi r^2) = \lambda^2/(4\pi).\ G_r.\ P_t.G_t/(4\pi r^2)$
  $\log P_r = \log P_t + \log G_t + \log G_r - (32.44 + 20 \log r + 20 \log f)$
  $L_{fs} = 32.44 + 20 \log r + 20 \log f$

Friis Free Space Path Loss Model



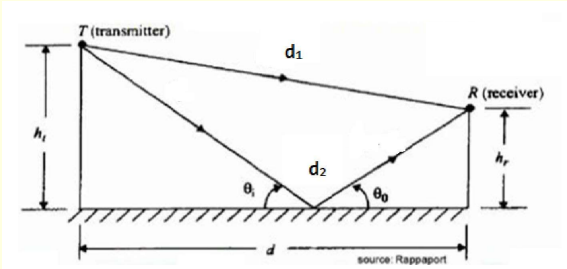
$P_r = P_t.G_t.G_r.\ (\lambda/(4\pi\, d))^2$

$$PL\,(\mathrm{dB}) = 10\log\frac{P_t}{P_r} = -10\log\left[\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2}\right]$$

- Friis free space model is valid for distance d in the far field which is a region beyond the far field distance $d_f = \frac{2D^2}{\lambda}$ where D is the largest linear distance of the antenna
- The received power can be estimated using a reference point beyound far field distance as follows:

$$P_r(d) = P_r(d_0)\left(\frac{d_0}{d}\right)^2 \qquad d \ge d_0 \ge d_f$$

2-Ray Path Loss Model



$$d_1 = \sqrt{(h_t - h_r)^2 + d^2}$$

$$d_1 = d\sqrt{\left(\frac{h_t - h_r}{d}\right)^2 + 1}$$

$$d_2 = \sqrt{(h_t + h_r)^2 + d^2}$$

$$d_2 = d\sqrt{\left(\frac{h_t + h_r}{d}\right)^2 + 1}$$

$$\Delta d = (d_2 - d_1) = d\left(\sqrt{\left(\frac{h_t+h_r}{d}\right)^2 + 1} - \sqrt{\left(\frac{h_t-h_r}{d}\right)^2 + 1}\right)$$

$$\approx d\left(\left(1 + \tfrac{1}{2}\left(\tfrac{h_t+h_r}{d}\right)^2\right) - \left(1 + \tfrac{1}{2}\left(\tfrac{h_t-h_r}{d}\right)^2\right)\right) \text{ applying Taylor expantion}$$

$$\approx \frac{2h_t h_r}{d}$$

$$\Delta\phi = \frac{2\pi\Delta d}{\lambda} \approx \frac{4\pi h_t h_r}{\lambda d}$$

$P_r = P_t.G_t.G_r.\ (\lambda/(4\pi\, d))^2$

$$P_r \approx P_t\left(\frac{\lambda\sqrt{G}}{4\pi d}\right)^2 \times |1 - e^{-j\Delta\phi}|^2 \approx P_t\left(\frac{\lambda\sqrt{G}}{4\pi d}\right)^2 \times |1 - (1 - j\Delta\phi)|^2 = P_t\left(\frac{\lambda\sqrt{G}}{4\pi d}\right)^2 \times \Delta\phi^2 = P_t\frac{G h_t^2 h_r^2}{d^4}$$

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4} \qquad d \gg \sqrt{h_t h_r}$$

Simplified Path Loss Model

Friis Free Space Model

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2}$$

2-Ray Model

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4} \qquad d \gg \sqrt{h_t h_r}$$

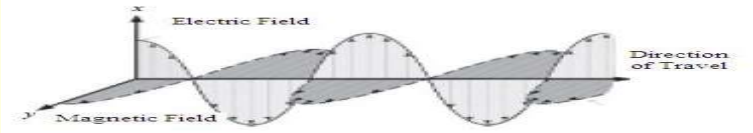
Simplified Path Loss Model

$$P_r = P_t K \left[\frac{d_r}{d}\right]^{\gamma}, \qquad 2 \le \gamma \le 8$$

- Used when the path loss is dominated by Reflections
- Most important parameter is the path loss exponent γ determine empirically

Propagation Mechanism

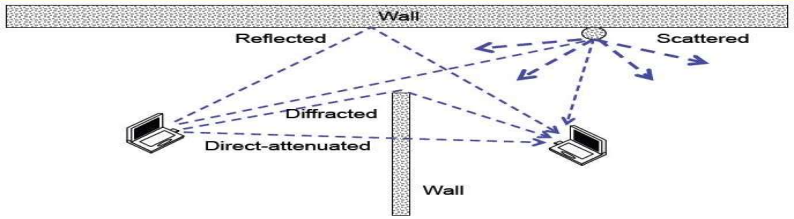
- Electromagnetic waves propagation



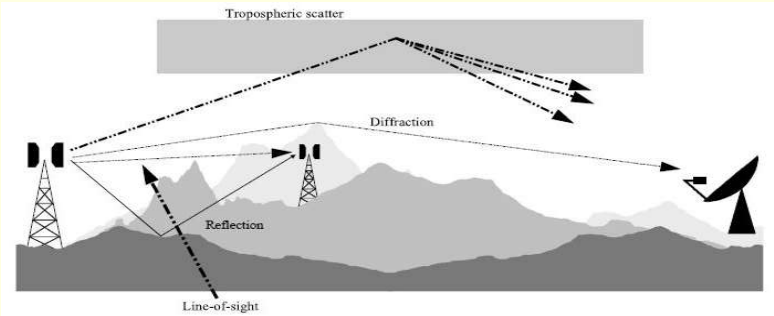
- Reflection:
  - Propagation wave encounters a surface which is large enough as compared to the wavelength of the signal.
- Diffraction:
  - Propagation wave encounters a surface edge which is large enough as compared to the wavelength of the signal.
- Scattering:
  - Propagation wave encounters an obstacle whose size is on the order of the wavelength of the signal or less.

Propagation Mechanism (Cont.)

- Indoor Propagation:



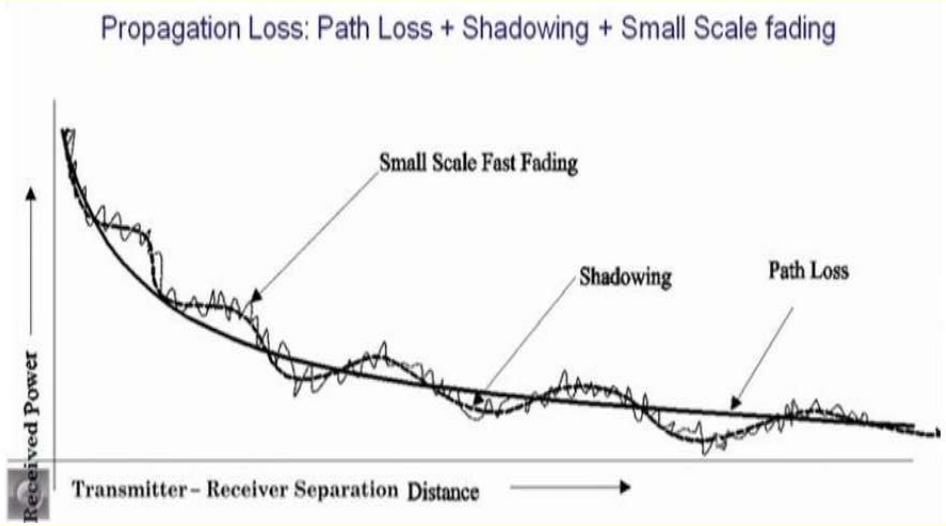
- Outdoor Propagation



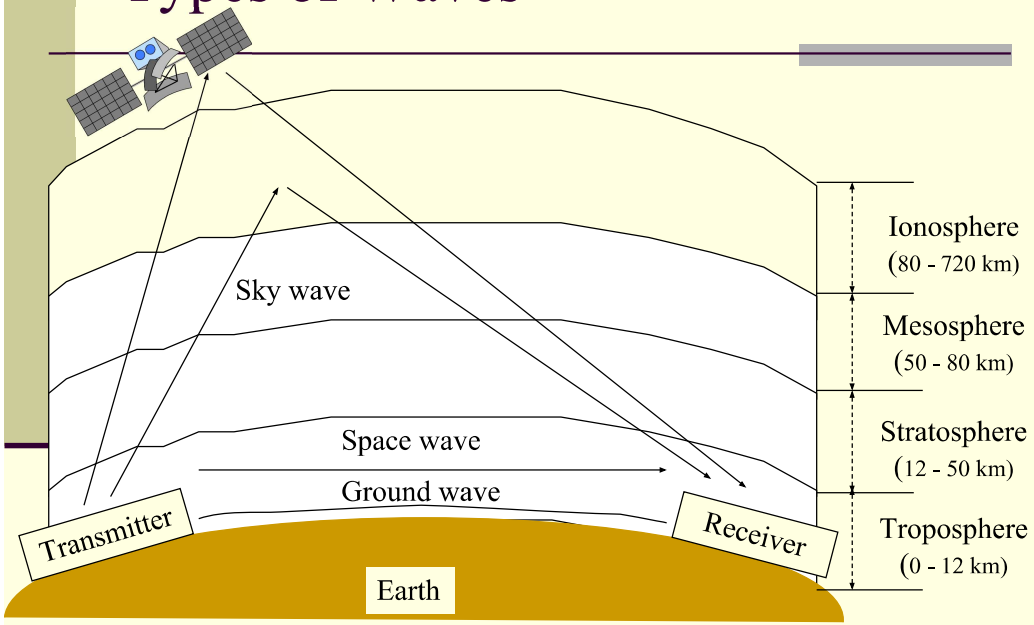
Effect of Multipath Propagation

- Fading:
  - Fading refers to the fluctuation in signal strength received at the receiver.
  - Due to multipath propagation, multiple copies of the same signal may arrive at the receiver with different amplitude, phase or delay. They may cause Intersymbol Interference(ISI).
  - ISI occurs sometimes constructivly and sometimes destructively resulting in the variation in power level of the received signal.
  - Rapid fluctuation in the signal strength refers to the Fast Fading.

Overall Propagation Loss



Types of Waves



Radio Frequency Bands

Classification Band	Initials	Frequency Range	Characteristics
Extremely low	ELF	< 300 Hz	Ground wave
Infra low	ILF	300 Hz - 3 kHz	
Very low	VLF	3 kHz - 30 kHz	
Low	LF	30 kHz - 300 kHz	
Medium	MF	300 kHz - 3 MHz	Ground/Sky wave
High	HF	3 MHz - 30 MHz	Sky wave
Very high	VHF	30 MHz - 300 MHz	Space wave
Ultra high	UHF	300 MHz - 3 GHz	
Super high	SHF	3 GHz - 30 GHz	
Extremely high	EHF	30 GHz - 300 GHz	
Tremendously high	THF	300 GHz - 3000 GHz	